

Sandbox Networks: Boosting user- generated content in gaming with powerful cloud infrastructure

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ABOUT SANDBOX



By leveraging the global network and custom virtual machines on Google Cloud, Sandbox Networks improves gaming experience on its platform dedicated to user-generated content for gamers around the world.

About Sandbox Networks

Sandbox Networks is a gaming startup founded in 2015 dedicated to building a global ecosystem for user-generated content. Its gaming platform currently hosts more than 4,000 games and has around 200 million users around the world. By supporting users to create games with quality IT infrastructure, Sandbox Networks operates a wide range of popular games, including Bedwars, Skyblock and Hero Tycoon.

Industries: Games

while reducing
infrastructure costs

Location: China

Storage (<https://cloud.google.com/storage/>),

Google Cloud results

- Reduces network latency by 10% with the comprehensive global network
- Increases gaming resource download speed by 10% with Cloud CDN
- Helps save 30% of infrastructure costs with custom VMs on Compute Engine

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About Baidao Data

Baidao Data is an industry-leading cloud management service provider (Google Cloud partner), headquartered in Hong Kong, with branches in Shenzhen, Shanghai, Beijing, Chengdu and other places. Baidao Data has a deep accumulation in public cloud management services, providing cloud and cloud integration services, including cloud consulting and design services, data migration services, operation and maintenance hosting services, big data and AI platform services, hybrid cloud management services, security services and integration services, etc., serve 200+ customers, covering finance, high-tech manufacturing, Internet, ecommerce, games, and other industries.

accumulation in public cloud management

Google Cloud (<https://cloud.google.com/>)

integration services, including cloud

Cloud CDN (<https://cloud.google.com/cdn/>)

Cloud Storage (<https://cloud.google.com/storage/>)

maintenance hosting services, big data

Compute Engine

(<https://cloud.google.com/compute/>)

and integration services, etc., serve 200+

User-generated content (UGC) is
(<https://venturebeat.com/games/hc-content-is-changing-the-game-industri-good/#:~:text=UGC%20is%20the%20support%20those%20creator%20cor>

customers, covering finance, high-tech manufacturing, Internet, ecommerce, games, and other industries.

in the games industry. With stor gamers themselves, UGC-based games are known for higher creativity and tighter social bonds among users. Alongside the growing number of UGC-based games, leading gaming studios are inspired to adopt similar strategies and release overwhelmingly successful games like Fortnite and Minecraft.

Sandbox Networks is on a mission to build a global ecosystem for UGC-based games. Founded in 2015, the gaming startup operates a gaming platform that currently hosts more than 4,000 games, which are mostly developed by external UGC creators. Having around 200 million users around the world, Sandbox Networks offers a wide range of mobile games from collaborative player versus player games like Bedwars to role-playing games like Hero Tycoon.

"We want to provide a platform where gamers can access various games at once and have the opportunity to release the games they create," explains Weiming Chen, CEO at Sandbox Networks.

"Our gaming platform is not only a place for users to have fun, but also to share creativity and inspirations."

To provide gaming services to users around the world, Sandbox Networks first used a public cloud platform to deploy all its games. However, the network connectivity was not ideal in regions with suboptimal internet infrastructure like India, which

occasionally caused connection disruptions between the end devices and Sandbox Networks's game servers. Furthermore, as the number of its games increased, Sandbox Networks needed more cost-effective IT infrastructure to maintain efficient operations. As a result, Sandbox Networks decided to migrate its cloud gaming platform to Google Cloud (<https://cloud.google.com/>) for better connectivity and lower infrastructure costs.

"To ensure quality gaming experiences in every region where we operate and maximize the value of our spending, we had to move our games to a global cloud platform with high connectivity and cost-effective infrastructure, and Google Cloud fulfills those criteria," says Hengguo Cheng, technology director at Sandbox Networks.

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Sandbox Networks*

Improving gaming experience with a highly connected global network

With the on-site technical support of the Google Cloud team, Sandbox Networks migrated all its games to Google Cloud in mid-2022 within only one month. Now, the startup leverages the global network of Google Cloud to support the data transfer of its games. Since Google Cloud offers more network nodes around the world, which leads to faster connection speed, Sandbox Networks is able to lower its gaming latency by 10%.

The connection quality of Sandbox Networks's games has also been improved. When using the previous cloud platform, the Sandbox Networks

team regularly received user reports complaining about not being able to connect to its game servers in regions with poorer network infrastructure. Since the migration, the team hasn't received any connection failure report.

On top of that, Sandbox Networks leverages Cloud CDN (<https://cloud.google.com/cdn>) to accelerate its gaming content delivery. By sending content closer to users through Cloud CDN, the gaming startup has increased the resource download speed and success rate of its games by 10% respectively.

"The comprehensive global network of Google Cloud and Cloud CDN have helped us significantly improve the gaming experiences that we offer. Now, users around the world can enjoy our games to the fullest without being bothered by network glitches," notes Cheng.

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Reducing infrastructure costs with custom VMs on Compute Engine

Sandbox Networks uses the virtual machines (VMs) on Compute Engine (<https://cloud.google.com/compute>) as servers to run its games, and Cloud Storage (<https://cloud.google.com/storage>) to store all the game resources. As Sandbox Networks operates a wide range of games, the computing resources required for each game vary greatly. With the custom VMs of Compute Engine, Sandbox Networks is able to change VM specifications according to its needs and avoid paying for excess computing resources. Combined with the committed use discounts offered by Google Cloud, Compute Engine has helped

Sandbox Networks reduce its IT infrastructure costs by 30%.

"The possibility of requesting computing resources according to actual use on Compute Engine is very beneficial to us, because we can use our budget more cost-effectively," says Cheng. "The VMs on Compute Engine are highly reliable. We've never encountered any errors since the migration to Google Cloud."

Supporting the UGC gaming ecosystem with quality services and features

Moving forward, Sandbox Networks will continue expanding its UGC-based gaming platform through providing quality services supported by the Google Cloud infrastructure. The Sandbox Networks team is also testing Translation AI

(<https://cloud.google.com/translate>) to facilitate communications between gamers who speak different languages.

Ping Kong, senior product manager at Sandbox Networks, says, "With the high-performance infrastructure of Google Cloud, we're able to provide more high quality games and attract more UGC creators. We're excited to see our UGC gaming ecosystem keep growing and fostering more original gaming content."

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